

Counterfactual validation of image-grounded P&ID tag reading

Source-disjoint
drawing IDs

Answer-hidden
scorer store

Actual budget
pixels / tokens

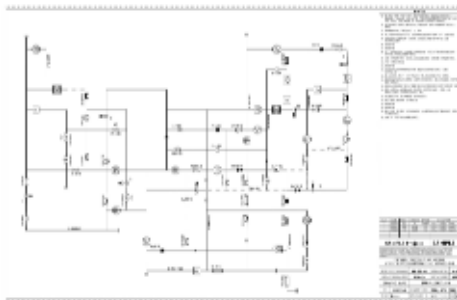
Counterfactual
correct / wrong / none

Task interval
source bootstrap

Deterministically selected complete-recovery case

Question: Find all tags of 27 symbols that begin with RO | Reference: ro-10 364, ro-10 525, ro-10 545

PIDQA sheet 282 (CC0)
complete source drawing; no answer-guided crop



16 eligible records; minimum SHA-256 rank 065e6468962f...

768 correct image

F1=0.00 | exact=0

Raw: R0101, R0102, R0103, R0104, R0105,
R0106, R0107, R0108, R0109, R0110,
R0111, R0112, R0113, R0114, R0115,
R0116, R0117, R0118, R0119, R0120,
R0121, R0122, R0123

Parsed: ro101, ro102, ro103, ro104, ro105, ro106,
ro107, ro108, ro109, ro110, ro111, ro112,
ro113, ro114, ro115, ro116, ro117, ro118,
ro119, ro120, ro121, ro122, ro123

3072 correct image

F1=1.00 | exact=1

Raw: R0-10 364, R0-10 545, R0-10 525

Parsed: ro-10 364, ro-10 525, ro-10 545

3072 source-shuffled image

F1=0.00 | exact=0

Raw: R0-10 517, R0-10 599, R0-10 969, R0-10
435, R0-10 979, R0-10 463, R0-10 351,
R0-10 327, R0-10 867, R0-10 193, R0-10
878, R0-10 372, R0-10 371, R0-10 878,
R0-10 372, R0-10 371, R0-10 878, R0-10
372, R0-10 371, R0-10 878, R0-10 372,
R0-1

Parsed: ro-1, ro-10 193, ro-10 327, ro-10 351,
ro-10 371, ro-10 372, ro-10 435, ro-10
463, ro-10 517, ro-10 599, ro-10 867,
ro-10 878, ro-10 969, ro-10 979

No image

F1=0.00 | exact=0

Raw: R0101, R0102, R0103, R0104, R0105,
R0106, R0107, R0108, R0109, R0110,
R0111, R0112, R0113, R0114, R0115,
R0116, R0117, R0118, R0119, R0120,
R0121, R0122, R0123, R0124, R0125,
R0126, R0127

Parsed: ro101, ro102, ro103, ro104, ro105, ro106,
ro107, ro108, ro109, ro110, ro111, ro112,
ro113, ro114, ro115, ro116, ro117, ro118,
ro119, ro120, ro121, ro122, ro123, ro124,
ro125, ro126, ro127